

BVAR Forecasting – Clean Notebook

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#####
# BVAR CLEAN NOTEBOOK - Victor Version (One-Block Code)
# - Same 3 variables Victor used
# - Minnesota priors
# - Lag p = 2
# - Holdout: last 24 observations
# - Outputs clean CSVs:
#   bvar_rmse_by_var.csv
#   bvar_holdout_summary.csv
#####

# ===== 0. Libraries =====
options(repos = c(CRAN = "https://cloud.r-project.org"))

if (!requireNamespace("librarian", quietly = TRUE)) install.packages("librarian")
librarian::shelf(readr, dplyr, lubridate, tidyr, BVAR)

## Warning: package 'readr' was built under R version 4.5.2

# ===== 1. Load Data =====
df <- readr::read_csv("balanced_can_md.csv", show_col_types = FALSE) |>
  dplyr::mutate(
    Date      = as.Date(Date),
    GDP_per_w = GDP_new - EMP_CAN
  ) |>
  dplyr::arrange(Date)

# Pick Victor's 3 small-VAR variables
# If your dataset already has infl_can, house_price, adjust the `select()` row accordingly
df_small <- df |>
  dplyr::select(
    Date,
    infl_can      = CPI_ALL_CAN,    # rename your inflation series here if needed
    gdp_pc_proxy  = GDP_per_w,
    house_price   = NHOUSE_P_CAN
  ) |>
  tidyr::drop_na()

Y_bvar <- as.matrix(df_small[, c("infl_can", "gdp_pc_proxy", "house_price")])
n_total <- nrow(Y_bvar)

# ===== 2. 24-step Holdout =====
h_out <- 24L
train_n <- n_total - h_out
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Y_train <- Y_bvar[1:train_n, , drop = FALSE]
Y_test  <- Y_bvar[(train_n + 1):n_total, , drop = FALSE]

# ===== 3. Fit BVAR (Vector-style) =====
p_bvar      <- 2L
priors_bvar <- BVAR::bv_priors()

set.seed(123)
bvar_model <- BVAR::bvar(
  data   = Y_train,
  lags   = p_bvar,
  n_draw = 8000L,
  n_burn = 4000L,
  priors = priors_bvar,
  verbose = FALSE
)

# ===== 4. Compute Holdout RMSE =====
rmse_vec <- BVAR::rmse(bvar_model, holdout = Y_test)

bvar_rmse_by_var <- data.frame(
  Variable = colnames(Y_bvar),
  RMSE     = as.numeric(rmse_vec)
)

# Average RMSE
rmse_avg <- mean(bvar_rmse_by_var$RMSE)

# ===== 5. Export Result Tables =====
readr::write_csv(bvar_rmse_by_var, "bvar_rmse_by_var.csv")

bvar_summary <- data.frame(
  Model      = "BVAR (3 variables, p=2)",
  Lag_p      = 2L,
  RMSE_1to12 = rmse_avg
)

readr::write_csv(bvar_summary, "bvar_holdout_summary.csv")

print("===== BVAR RESULTS =====")

## [1] "===== BVAR RESULTS ====="

print(bvar_rmse_by_var)

##      Variable      RMSE
## 1   infl_can 0.003549497
## 2  gdp_pc_proxy 0.002551991
## 3  house_price 0.002836908

print(bvar_summary)

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##                               Model Lag_p  RMSE_1to12
## 1 BVAR (3 variables, p=2)      2 0.002979465
```

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print("CSV files saved:")
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print(" - bvar_rmse_by_var.csv")
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